

Model Version and Documentation Record

Churn Prediction Model — CRM Scope

1. Purpose of This Document

This document records the version details of the churn prediction model so it can be tracked, reused, and audited later. It is meant to be a simple production record — not a technical deep-dive.

2. Model Identification

- **Model Name:** Churn Prediction Model
- **Scope:** CRM Scope — Patient Support Program (PSP)
- **Version:** v1.0
- **Model Type:** XGBoost Classifier
- **Status:** Proof of Concept (hackathon build)
- **Owner / Team:** ML Team, Cognizant PSP Hackathon Project

3. What the Model Was Trained On

The model was trained using the CRM-side synthetic dataset (Patient, Program Enrollment, Patient Support Interaction, Eligibility Population, and Patient Outcome Status tables). Details of every input feature are covered in the separate Final Feature Definitions document.

4. Performance Summary

- **Accuracy:** 75 percent on held-out test data
- **ROC AUC:** 0.83

Full performance details are available in the Churn Prediction Model Documentation shared earlier.

5. Stored Artifact

- **File Name:** churn_prediction_model.pkl
- **Contents:** Trained model, category encoders, and the exact feature order needed to run predictions

6. Version History

Version	Date	Change Description	Author
v1.0	18 Aug 2026	Initial churn prediction model built and validated on synthetic CRM data.	ML Team

7. Notes

This is the first version of the model, built for the hackathon proof of concept. Any future retraining or changes must follow the Model Update and Versioning Procedure document, and this version history table should be updated accordingly.